by their mass and charge. Their mass is 4 atomic mass units, and they carry two elementary charges of electricity. We express this shortly by saying that their *mass number* is 4, and their atomic number is 2. Accordingly, the helium atom is represented by the symbol $_2He^4$, where the superscript denotes the mass number and the subscript the atomic number. This kind of symbolism is applied to the atoms of all the elements. When an alpha particle is ejected by an atomic nucleus, it carries with it not only its own mass, but also its own charge. The nucleus loses both this mass and charge; its atomic number (the number of elementary quanta of electricity present on it) is reduced by 2, whereas its mass number is reduced by 4. On the other hand, a beta particle is a negative electron. Its mass number is approximately 0, and its atomic number is −1. Accordingly, if e stands for ‘electron’ in general, the symbol of the negative electron may be written as $_{-1}e^0$. Thus, when an electron is emitted, the mass of the nucleus remains practically unchanged, whereas its positive charge increases by 1, due to the loss of one elementary quantum of negative electricity. Hence, both the emission of an alpha particle and the emission of a beta particle result in a change in the atomic number. Since the chemical properties of the elements are determined by the nuclear charge number, the atomic number, it is evident that both alpha and beta radiation must result in a transmutation of elements.

Let us now examine these facts more closely, using radium as our example. The mass number of radium is 226, its atomic number is 88, and consequently, its symbol is $_{88}Ra^{226}$. The radium atom contains 88 extranuclear electrons; since 86 electrons form completed shells, the chemical properties of radium are determined largely by the two electrons which revolve around the nucleus in the outermost, incomplete shell. Therefore, the chemical properties of radium are similar to those of one of the alkaline earth metals, such as barium or strontium. The radium nucleus is an emitter of alpha rays, and as a result of this radiation, its mass is reduced to 222, and its nuclear charge, its atomic number, to 86. A new element—likewise radioactive—is formed: the inert gas radon, the symbol of which is $_{86}Rn^{222}$. Due to its nuclear charge number, the radon atom contains only 86 electrons, which are arranged in completed shells. This atom is therefore chemically inactive—this element is an inert gas. The process of the emission of an alpha particle by the radium atom and the formation of a radon atom are indicated by the following formula:

$$_{88}Ra^{226} \rightarrow {}_{86}Rn^{222} + {}_{2}He^{4}$$

The symbol to the left of the arrow is that of the radiating atom; the